

SEG800 Image Processing Cheatsheet

Based on Module 7 and Module 8 study summaries

Concept Study Notes

Concept	Reminder
Digital image processing	Processing a digital image using a digital computer.
Human vision: cones	Cones handle bright-light and color vision.
Enhancement vs restoration	Enhancement is subjective; restoration uses mathematical/probabilistic models.
Affine transformations	Preserve parallelism.
Quantization	Converts continuous brightness values into discrete levels.
Weber ratio	Measures brightness discrimination.
Digitizer	Converts analog signals into digital form.
Image energy source	Most digital images use the electromagnetic spectrum.
Point operation	Depends only on the corresponding input pixel.
Negative transformation	For 8-bit images: $s = 255 - z$.
Neighborhood operation	Image smoothing uses neighboring pixels.
Geometric transformation	Changes pixel locations.
Histogram equalization	Enhances contrast by spreading/flattening the histogram.
Gamma correction	Mainly changes brightness/intensity perception.
Translation	$u = x + t_x, v = y + t_y$.

Core Definitions

Image: a 2D function $f(x, y)$, where x, y are spatial coordinates and $f(x, y)$ is intensity or gray level.

Digital image: an image where x, y , and intensity values are finite and discrete.

Pixel: one image element with location (x, y) and intensity $f(x, y)$.

Image enhancement: improves an image for a specific application; subjective.

Image restoration: improves an image using objective mathematical/probabilistic degradation models.

Sampling: converts continuous spatial coordinates into discrete coordinates; related to spatial resolution.

Quantization: converts continuous brightness/intensity values into discrete digital values; related to intensity resolution.

Point operation: each output pixel depends only on the same input pixel.

$$I_{out}(i, j) = f(I_{in}(i, j))$$

Neighborhood operation: output depends on a pixel and its neighbors. Example: smoothing/blurring.

Geometric transformation: maps (x, y) to (u, v) ; changes locations, not intensity values.

Affine transformation: preserves parallelism. Examples: translation, scaling, rotation, shear.

Human Vision

Cones: bright-light/photopic vision; sensitive to color.

Rods: dim-light/scotopic vision; sensitive to low illumination.

Blind spot: area with optic nerve and no receptors.

Formula Box

Weber ratio

$$\frac{\Delta I_c}{I}$$

ΔI_c is the minimum detectable change in illumination; I is background illumination.

Brightness levels

$$2^b$$

b is number of bits. Example: 8 bits gives $2^8 = 256$ levels.

Negative transformation

$$s = 255 - z$$

z is input intensity; s is output intensity.

Gamma correction

$$s = cr^\gamma$$

r is input intensity, s is output intensity, c is a constant, and γ is gamma.

Lens formula

$$\frac{1}{z_o} + \frac{1}{z_i} = \frac{1}{f}$$

z_o is object distance, z_i is image distance, and f is focal length.

Histogram

$h(r_k)$ = number of pixels with intensity r_k

Normalized histogram

$$p(r_k) = \frac{h(r_k)}{MN}$$

Histogram equalization

$$s_k = T(r_k) = (L - 1) \sum_{j=0}^k p(r_j)$$

L is the number of intensity levels.

Geometric Transform Formulas

Homogeneous coordinate form:

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Translation

$$u = x + t_x, \quad v = y + t_y$$

$$\begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix}$$

Scaling

$$u = s_x x, \quad v = s_y y$$

$$\begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Rotation, counterclockwise

$$u = x \cos \theta - y \sin \theta, \quad v = x \sin \theta + y \cos \theta$$

$$\begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Shear

$$u = x + y sh_x, \quad v = y + x sh_y$$

$$\begin{bmatrix} 1 & sh_x & 0 \\ sh_y & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Pixel Adjacency

For $p = (x, y)$:

$$N_4(p) = \{(x+1, y), (x-1, y), (x, y+1), (x, y-1)\}$$

$$N_D(p) = \{(x+1, y+1), (x+1, y-1), (x-1, y+1), (x-1, y-1)\}$$

$$N_8(p) = N_4(p) \cup N_D(p)$$

Two pixels p and q are **m-adjacent** if either:

$$q \in N_4(p)$$

or $q \in N_D(p)$ and $N_4(p) \cap N_4(q)$ has no pixels with values from V .

Checklist: confirm both pixels are in foreground set V ; check 4-neighbor first; for diagonal pixels, inspect common 4-neighbors.

Optics / Lens Formula

$$\frac{1}{z_o} + \frac{1}{z_i} = \frac{1}{f}$$

Solving for object distance:

$$\frac{1}{z_o} = \frac{1}{f} - \frac{1}{z_i}$$

$$z_o = \frac{1}{\frac{1}{f} - \frac{1}{z_i}}$$

Keep all distances in the same unit before calculating. Convert at the end if needed.

Homography Composition

If applying scale → rotate → translate, multiply in reverse application order:

H = TRS

Apply using:

[u; v; 1] = H [x; y; 1]

Geometric transformations move pixel locations but do not directly change pixel intensity values. Raster images may need interpolation or rounding.

Extra Study Topics

Processing levels

Level	Input	Output	Examples
Image processing	Image	Image	Noise reduction, contrast enhancement, sharpening
Image analysis	Image	Features	Edges, contours, objects
Computer vision	Image	Meaning	Recognition, scene understanding

Image acquisition pipeline

Energy source → Scene/object → Imaging system
→ Image plane → Digitized image

Most images use the electromagnetic spectrum: visible light, X-ray, infrared, ultraviolet, microwave, and radio waves. Other sources include acoustic waves, ultrasonic waves, electron beams, and synthetic images.

Digital image types

Type	Meaning
Binary	2D array of 0s and 1s
Grayscale	2D array of brightness values, often 0–255 or normalized 0–1
Color	Usually three 2D arrays, such as R, G, B
Multispectral	Includes frequencies beyond visible light
Depth map	Stores distance/depth information

Brightness adaptation: the eye changes sensitivity depending on current light level. It adapts to a large intensity range, but not all at once.

Weber ratio interpretation

Large Weber ratio → poor brightness discrimination
Small Weber ratio → better brightness discrimination

Visual perception effects: Mach band effect, simultaneous contrast, optical illusions.

Saturation: highest intensity value beyond which values are clipped.

Noise: random unwanted variation in image intensity.

Region terms: a path is a sequence of connected pixels; a connected set/region has paths between pixels inside it; a boundary/contour contains pixels adjacent to outside pixels.

Distance measures for p = (x, y) and q = (u, v):

De(p, q) = sqrt((x - u)^2 + (y - v)^2)
D4(p, q) = |x - u| + |y - v|

D8(p, q) = max(|x - u|, |y - v|)

Linear operation property

H[af1(x, y) + bf2(x, y)] = aH[f1(x, y)] + bH[f2(x, y)]

Image arithmetic

s = f + g, d = f - g, p = f x g, v = f/g

Linear blend / weighted addition

dst = α img1 + β img2 + γ

α and β are image weights; γ is a brightness offset.

Brightness and contrast point operations

Iout(i, j) = Iin(i, j) + b
Iout(i, j) = aIin(i, j)

Adding changes brightness; multiplying changes contrast.

Contrast stretching: expands intensity values to improve contrast.

Thresholding: separates pixels into dark/light groups.

pixel ≥ threshold → white, pixel < threshold → black

Gamma rule

γ < 1 → brighter, γ > 1 → darker

Transformation families

Transformation	Preserves
Translation	Orientation
Rigid / Euclidean	Lengths
Similarity	Angles
Affine	Parallelism
Projective	Straight lines; parallel lines may not stay parallel

Interpolation: assigns intensity when transformed locations are non-integer.

Type	Meaning
Nearest-neighbor	Closest pixel; fast but blocky
Bilinear	Nearby pixels; smoother
Bicubic	More neighbors; smoother but slower

Quick Memory Tips

- Enhancement = subjective; restoration = model-based.
- Cones = color/bright light; rods = dim light.
- Sampling = spatial coordinates; quantization = intensity values.
- Point operation = same pixel only; neighborhood operation = pixel plus neighbors.
- Geometric transformation = changes locations; pixel value stays the same.
- Affine preserves parallelism; projective preserves straight lines.
- Histogram equalization = contrast enhancement.
- Gamma correction = brightness perception; γ < 1 brighter, γ > 1 darker.
- Negative image: s = 255 - z.
- Translation: add tx and ty.
- Large Weber ratio = poor brightness discrimination; small Weber ratio = better discrimination.